

Early-Cycle Based Hybrid Neuro-Fuzzy Model for Lithium-Ion Battery SOH Prediction

ABSTRACT:

Lithium-ion batteries gradually lose their ability to store charge during use. This is called capacity fade. If we know how much health the battery has left which is called state of health or SOH, we can predict when the battery will go bad. This will be very useful especially in electric vehicles. Therefore, we need to accurately predict SOH as the battery ages. Battery failure is very complex. It changes slowly over time and there are small fluctuations.

LSTM and ANFIS are used to tackle both problems. LSTM is good at learning patterns over time. Therefore, it predicts an overall trend of reduction in SOH in different cycles. But this may lead to minor errors. ANFIS is good at nonlinear adjustment. So, we use ANFIS to find the error (residual) between the following: Actual SOH – LSTM estimated SOH. So ultimately LSTM predicts SOH first and ANFIS predicts the required correction. Final SOH = LSTM prediction + ANFIS correction. The dataset used is NASA's Prognostics Centre's Battery B0005.

The main result of this project is that the model predicts SOH more accurately than using LSTM or ANFIS alone. It captures both long-term degradation trends and small fluctuations in battery behaviour. In this regard, the model has low RMSE values, proving better performance in prediction. Thus, it may provide suitable real-time monitoring of battery health for electric vehicles.

Keywords: Lithium-ion battery, State of Health (SOH), LSTM, ANFIS, Battery degradation, Hybrid prediction model.

INTRODUCTION:

Electric vehicles, portable electronics, solar storage systems and backup power units use lithium-ion batteries.

They have wide applications due to long cycle life, and efficiency. However, because of repeated charge and discharge cycles that a battery must go through, performance gradually deteriorates. This process of deterioration results in a reduction in the available capacity, commonly referred to as the State of Health (SOH) of the battery.

Major reasons why SOH accurate approximation is required:

1. In order to avoid sudden failure of the batteries during operations.
2. To plan maintenance or cell replacement at the right time.
3. To ensure safe and efficient usage of the battery.
4. To extend battery life through the monitoring of degradation trends.

A traditional method for the estimation of SOH depends complex electrochemical models. These usually require expensive sensors that are very difficult to carry out in real-time. Also, aging is nonlinear due to which it doesn't follow any straight pattern through which any equation could predict.

To overcome all these, data-driven methods have recently become most important. They give predictions without physical battery equations, learning directly from real data on batteries. In particular:

Long Short-Term Memory (LSTM) networks can learn time-based trends in data very well, such as capacity variation in successive cycles.

ANFIS is an adaptive neuro-fuzzy inference system, which integrates neural networks with fuzzy logic to model uncertain or nonlinear relationships.

In this project, a hybrid SOH prediction system combines both of these models. The LSTM will be trained on the early degradation data of the battery and predicts the SOH for the further cycles. Further, the residual difference between the predicted and actual SOH is taken by ANFIS and corrects it according to voltage-based battery behaviour. This combined approach uses strengths from both models:

- LSTM captures the long-term capacity fading trends.
- ANFIS smooths nonlinear variations and noise in the data.

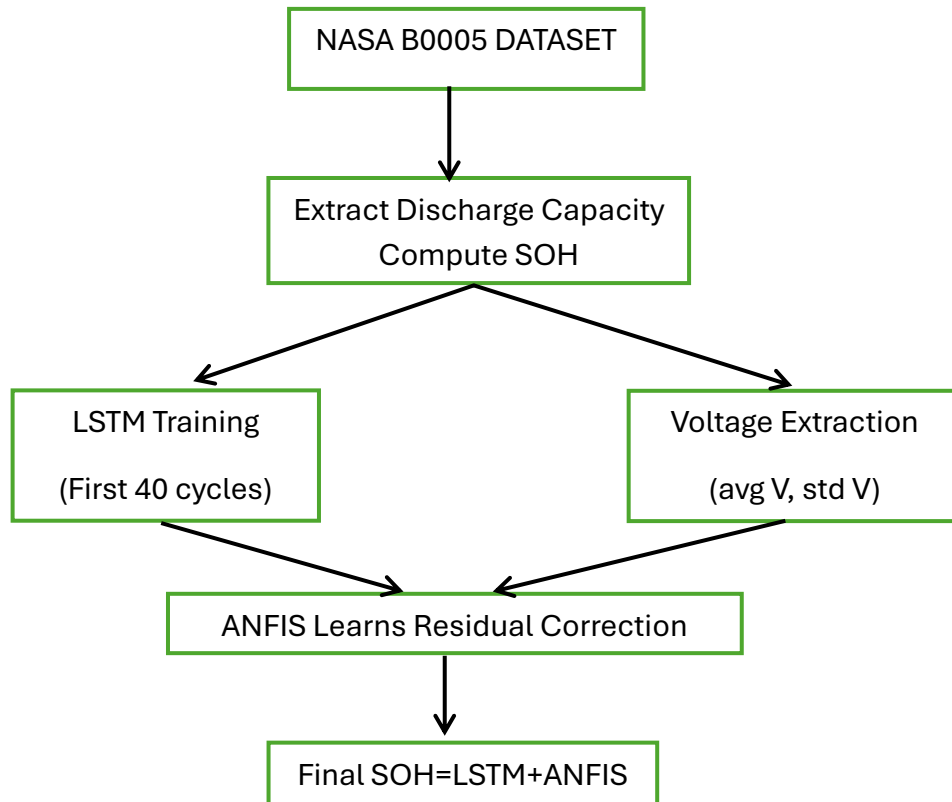
This approach of combining both ANFIS and LSTM gives more accurate result than by ANFIS or LSTM alone.

METHODOLOGY:

The proposed framework for SOH prediction consists of three major stages:

1. Data extraction and SOH computation
2. Early-cycle SOH trend learning using LSTM
3. Residual correction using ANFIS which gives final hybrid prediction output

Block diagram for this method is given below:



Dataset Information:

This approach was done using NASA B0005 Dataset. For each discharge cycle, the capacity value was extracted. The State of Health (SOH) was calculated by:

$$\text{SOH} = C_{\text{cycle}} / C_{\text{initial}}$$

This gives a SOH curve which decreases gradually with cycle count.

LSTM-Based Early Cycle Prediction:

Only the first 40 cycles were used for training the LSTM . This shows the real-world conditions as full lifespan data will not be available.

Input to LSTM:

- A window of 5 previous SOH values is given.
- It predicts the next cycle SOH.

After training, the LSTM model extrapolates SOH across the entire life cycle.

Voltage Extraction for ANFIS:

These features were extracted from each discharge cycle:

- Avg V -Average discharge voltage
- Std V -Standard deviation of discharge voltage

They reflect the aging condition of battery.

ANFIS-Based Residual Correction:

The difference between LSTM-predicted SOH and actual SOH was taken as residual:

$$\text{Residual} = \text{SOH}_{\text{actual}} - \text{SOH}_{\text{LSTM}}$$

ANFIS learns this residual:

(avg V, std V) → Residual

This can allow ANFIS to correct the errors in LSTM predictions.

Hybrid SOH Estimation:

The corrected final prediction is given as:

$$\text{SOH}_{\text{final}} = \text{SOH}_{\text{LSTM}} + \text{Residual}_{\text{ANFIS}}$$

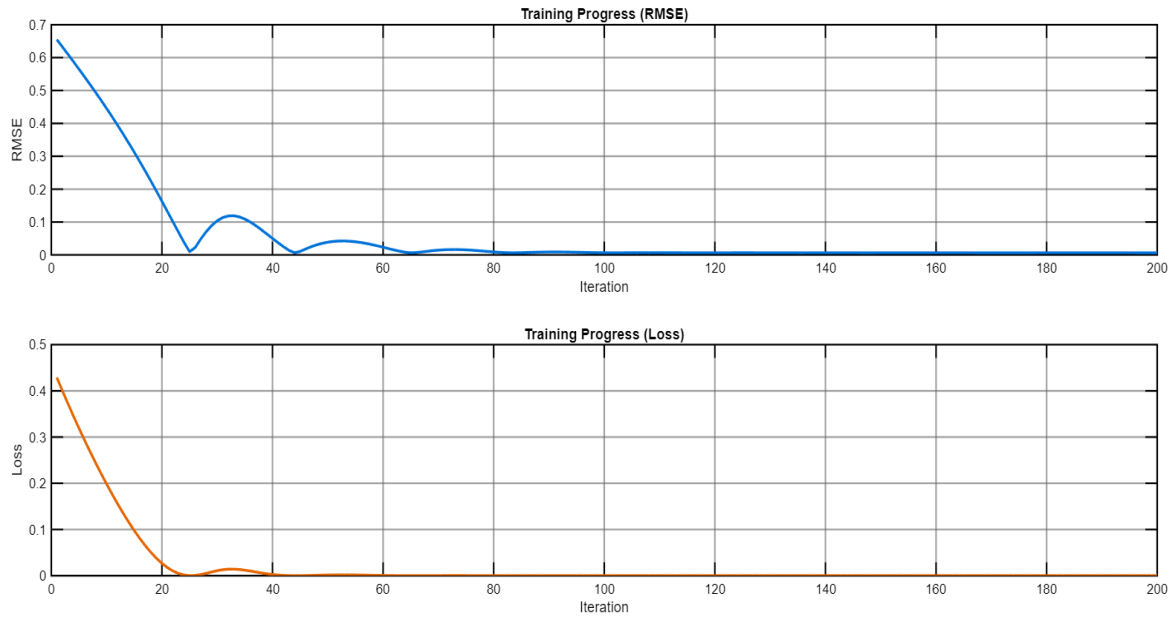
This fusion approach smooths the prediction curve and improves accuracy across the entire battery life cycle.

RESULTS AND ANALYSIS:

The hybrid LSTM–ANFIS model for SOH prediction was executed using the NASA B0005 lithium-ion battery dataset. The results are shown under three stages: 1) LSTM learning behavior, 2) ANFIS training and validation, and 3) final hybrid prediction performance.

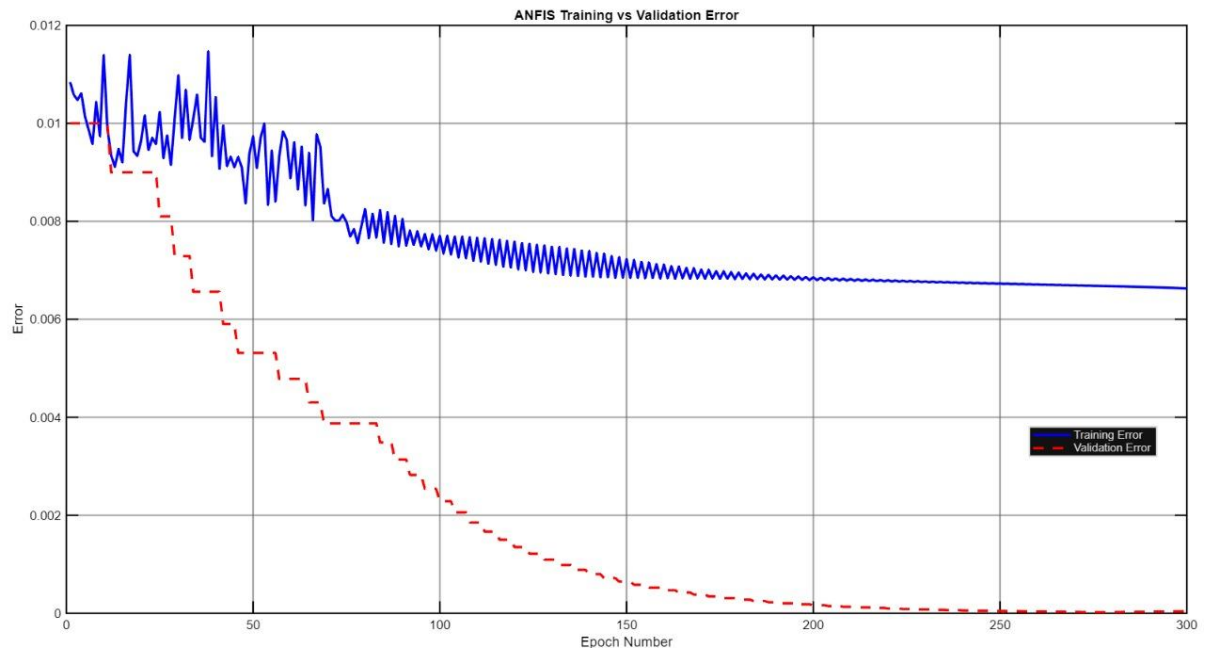
Training Behavior of LSTM:

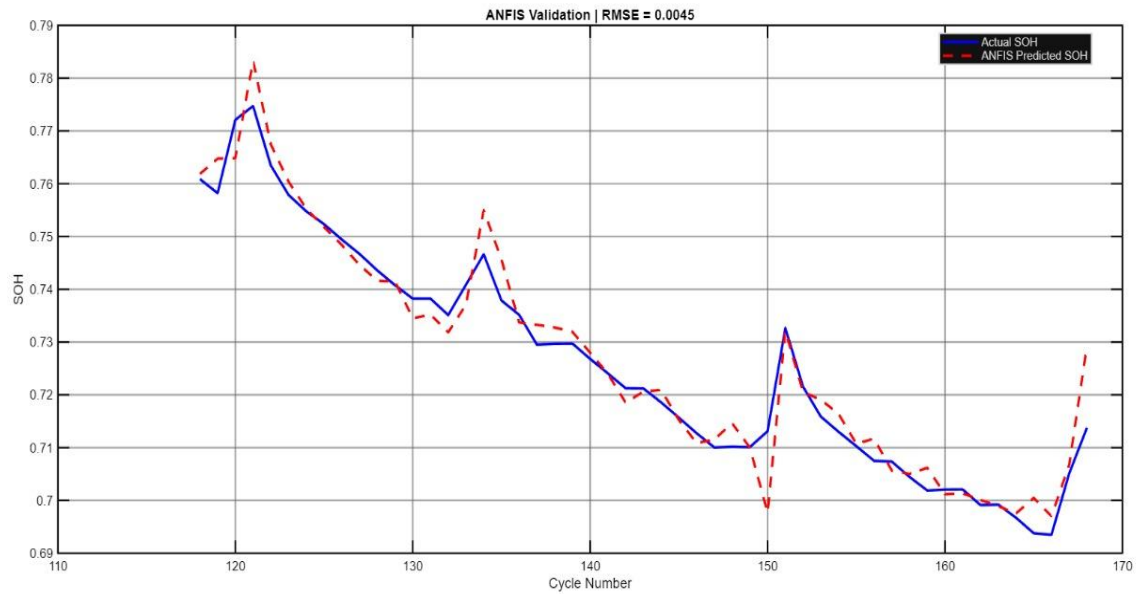
The LSTM network was trained using the first 40 cycles, where the SOH decreases gradually. In training progress minimum loss was observed. This says that the LTSM has learned the pattern successfully. However, during few cycles, the LSTM shows a drift due to the nonlinear behavior of battery aging. This drift is corrected by ANFIS.



ANFIS Training Performance:

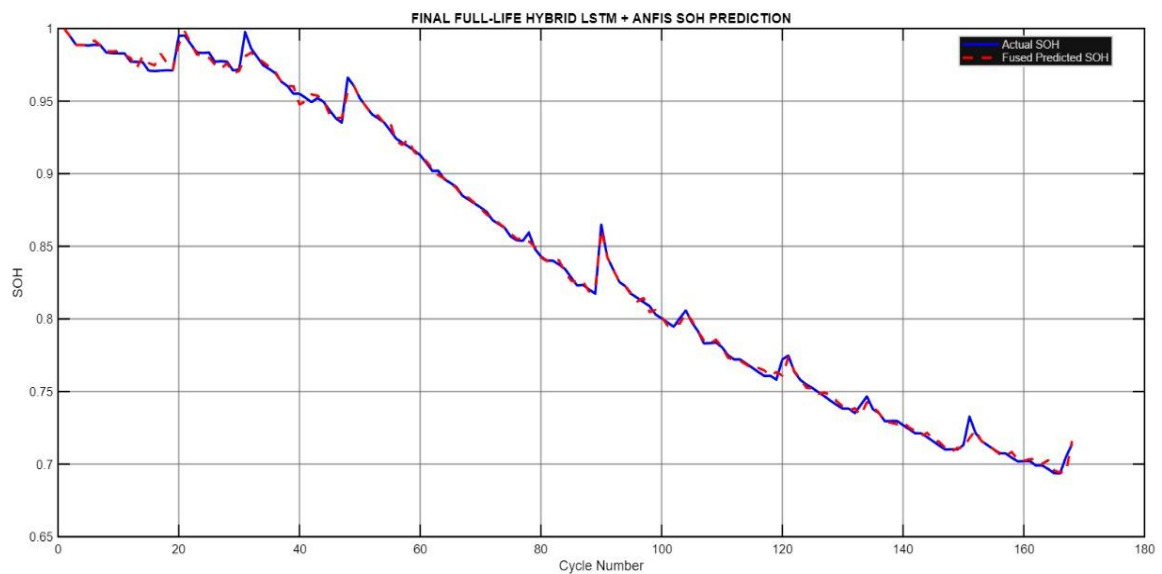
The ANFIS was trained with voltage features, such as avg V and std V, and residuals between actual and LSTM-predicted SOH. The training curve showed a progressive decrease of the training error with a validation curve that falls and did not overfit. ANFIS understood the non-linear behavior and even corrected errors from LSTM.





Final Hybrid Model Performance:

Finally, the outputs from LSTM and ANFIS were added to get the final SOH prediction. The SOH curve of the hybrid model closely matches with the actual SOH, which is better than LSTM or ANFIS alone.



Summary of results:

- LSTM trains to a good extent with the early data.
- ANFIS corrects the nonlinear deviations using voltage health indicators.
- The hybrid model produces improved long-term accuracy and stability.
- The approach applied will be useful for real-time BMSs.

CONCLUSION AND FUTURE WORK:

This paper focuses on the design and evaluation a SOH prediction model using a hybrid approach combining the Long Short-Term Memory network and Adaptive Neuro-Fuzzy Inference System methods, trained on the NASA B0005 lithium-ion dataset. The proposed SOH model initially demonstrates good tracking of early degradation while training using capacity data obtained during early cycles; however, its performance deteriorates as the nonlinear nature of the battery behavior becomes more pronounced with cycling. Thereafter, ANFIS is used to model and correct the residual errors with the inclusion of voltage features. The final prediction of this hybrid methodology, after incorporating ANFIS corrections with LSTM predictions, results in good agreement with the actual SOH curve over the full lifetime.

The results show that the proposed hybrid model performs well as compared to only LSTMs or ANFIS based models. The ability of LSTM to learn aging trends over a period of time, and the proficiency of ANFIS for managing nonlinear variations is found to be very effective in long-term SOH estimation. This model will be very useful for online health monitoring of batteries, which can easily be implemented using Battery Management Systems utilized in electric vehicles and stationary energy storage systems.

Future Work:

Possible future enhancements include:

1. Train the model with several varieties of batteries to improve generalization.
2. Add the following features: temperature, charge rate, and depth of discharge.
3. Deep temporal learning using advanced architectures including Transformer networks.
4. Online learning can be implemented so the model updates itself during battery usage.

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